

EfficientNet-B0 Cyclone Wind-Speed Prediction Model Training & Performance Report

Cyclone-X / TCAPS — Image-Based Intensity Analysis
Final evaluation: 1,533 held-out test images from 73 storms

1. Executive Summary

B0 is a four-channel satellite-image wind-speed regression model. It predicts maximum sustained wind speed in knots, then maps the prediction to Weak, Moderate, Strong or Very Strong. Final test performance: MAE 9.8299 kt, RMSE 13.4467 kt, R^2 0.7737, bias -0.0306 kt. Derived accuracy is 68.56% and Macro-F1 is 0.6324.

2. Dataset and Experimental Setup

10,538 genuine samples after duplicate removal; $128 \times 128 \times 4$ images; IR, WV, VIS and PMW channels; 485 storms. Train/validation/test = 7,577/1,428/1,533 images from 339/73/73 storms with no storm overlap.

3. Data Preprocessing

Invalid PMW values were replaced using the training PMW mean. Train-only normalization was used: IR 179.8942 ± 65.3210 ; WV 174.1156 ± 52.1954 ; VIS 65.9481 ± 61.3502 ; PMW 8.8655 ± 26.3539 .

4. Prediction Pipeline

Four-channel satellite image → PMW repair → train-derived normalization → EfficientNet-B0 feature extraction → adapted 4-channel first layer → $1280 \rightarrow 128 \rightarrow 1$ regression head → wind-speed prediction in kt → optional intensity mapping.

5. Training

Pretrained EfficientNet-B0; Huber loss; AdamW; 5-epoch head warm-up followed by full fine-tuning; initial backbone/head LR $1e-4/1e-3$; ReduceLROnPlateau; gradient clipping 1.0; early stopping patience 7; best validation checkpoint Epoch 32.

6. Training Behaviour

Validation MAE reached 10.3537 kt at Epoch 32. Training error continued decreasing later while validation plateaued, indicating a generalization gap. Training stopped at Epoch 39 after seven epochs without improvement.

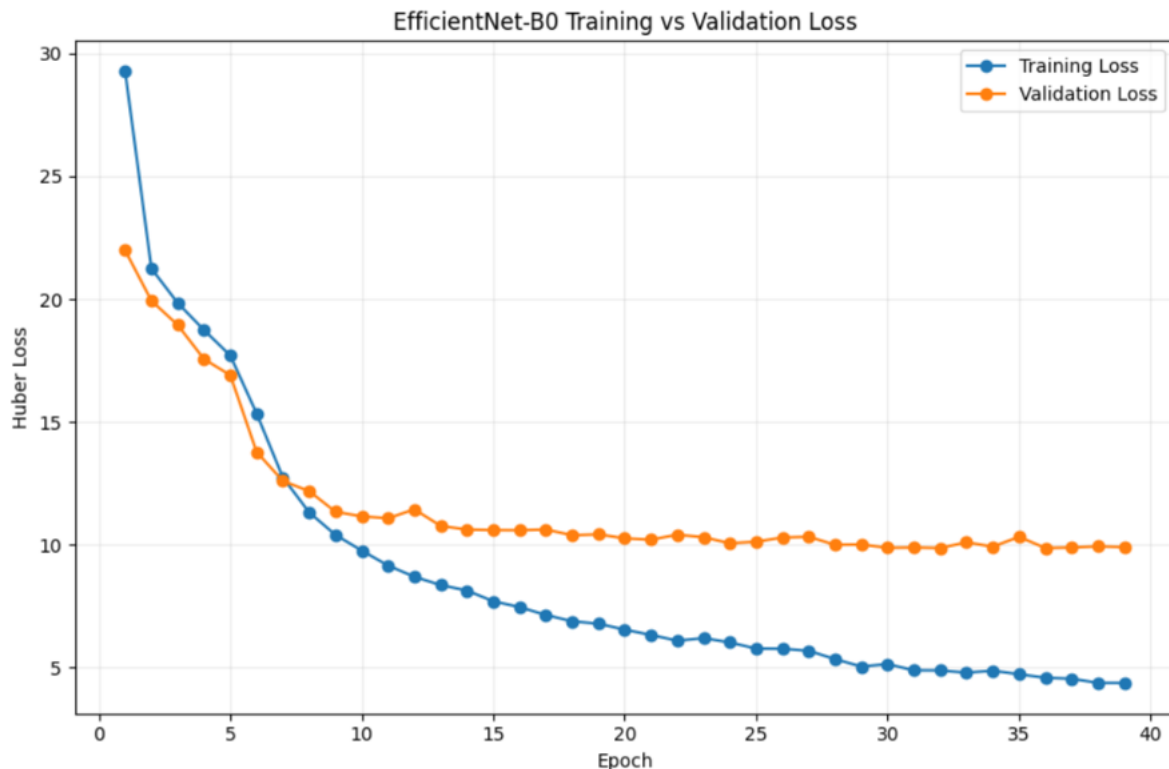


Figure 1. Training vs. validation loss.

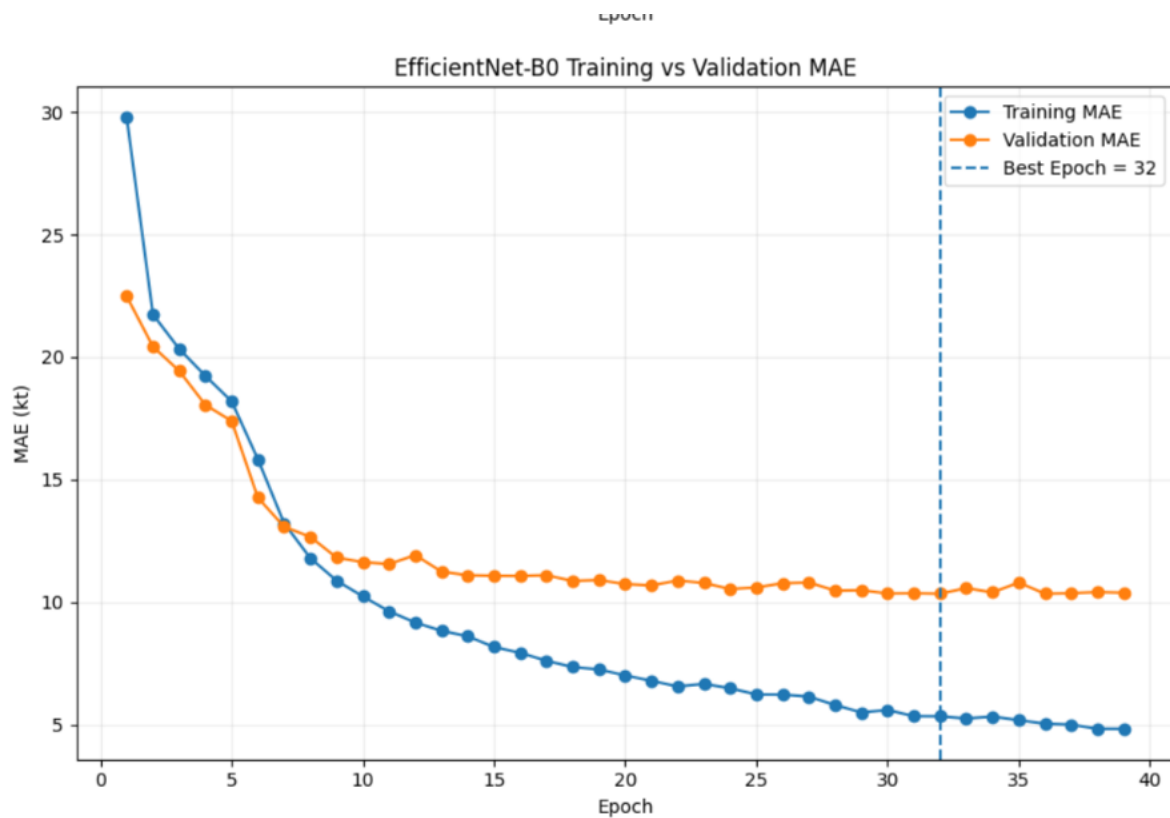


Figure 2. Training vs. validation MAE; best checkpoint Epoch 32.

7. Final Test Regression Performance

Metric	Result	Interpretation
MAE	9.8299 kt	Average absolute error
RMSE	13.4467 kt	Emphasizes large errors
R ²	0.7737	About 77% variance explained
Bias	-0.0306 kt	Essentially zero global bias

These regression metrics are primary because wind speed is the direct model output. Near-zero bias indicates no meaningful overall overprediction or underprediction.

8. Actual vs. Predicted Wind Speed

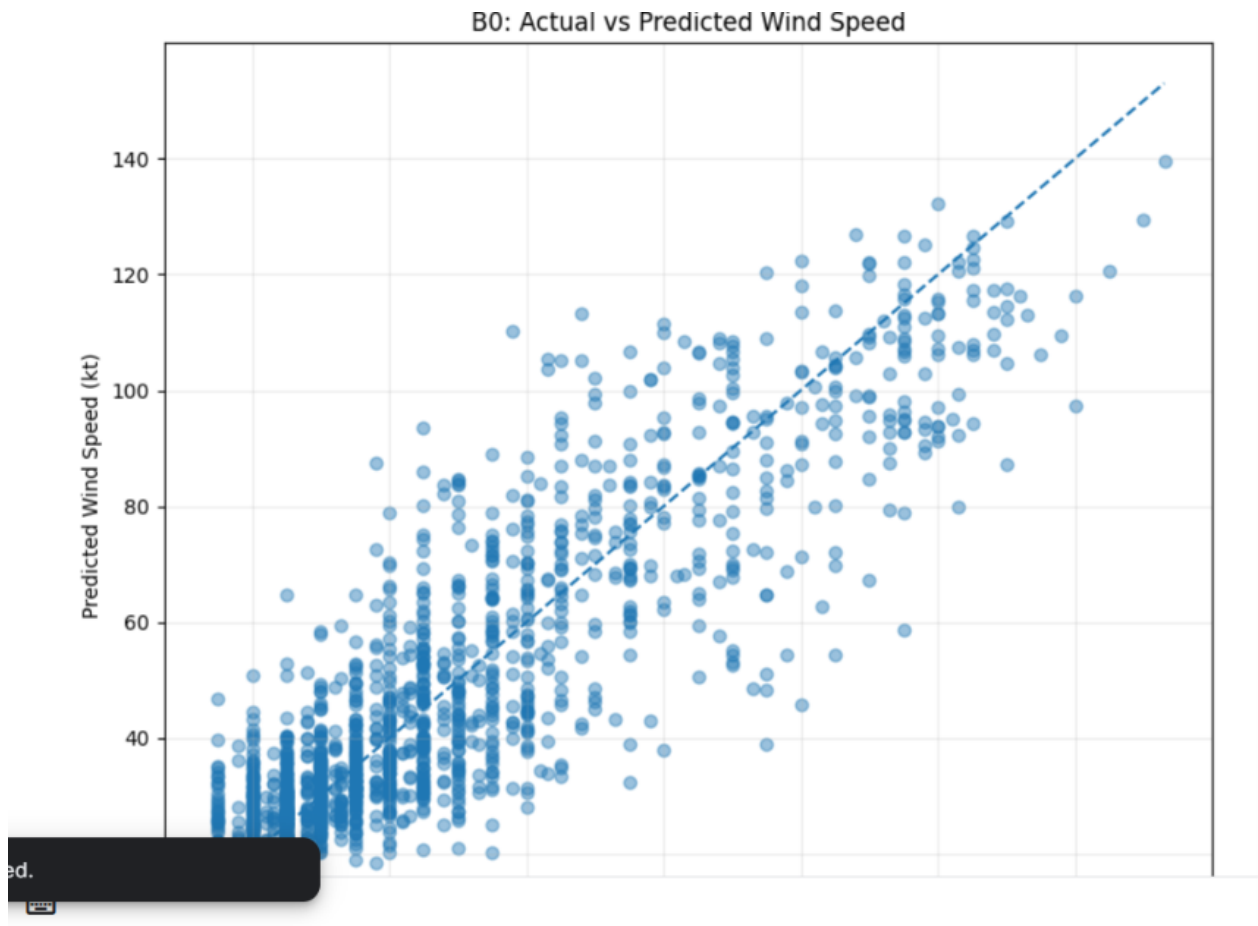


Figure 3. The positive trend around the perfect-prediction diagonal shows that B0 learned a meaningful image-to-wind relationship.

9. Residual / Error Analysis

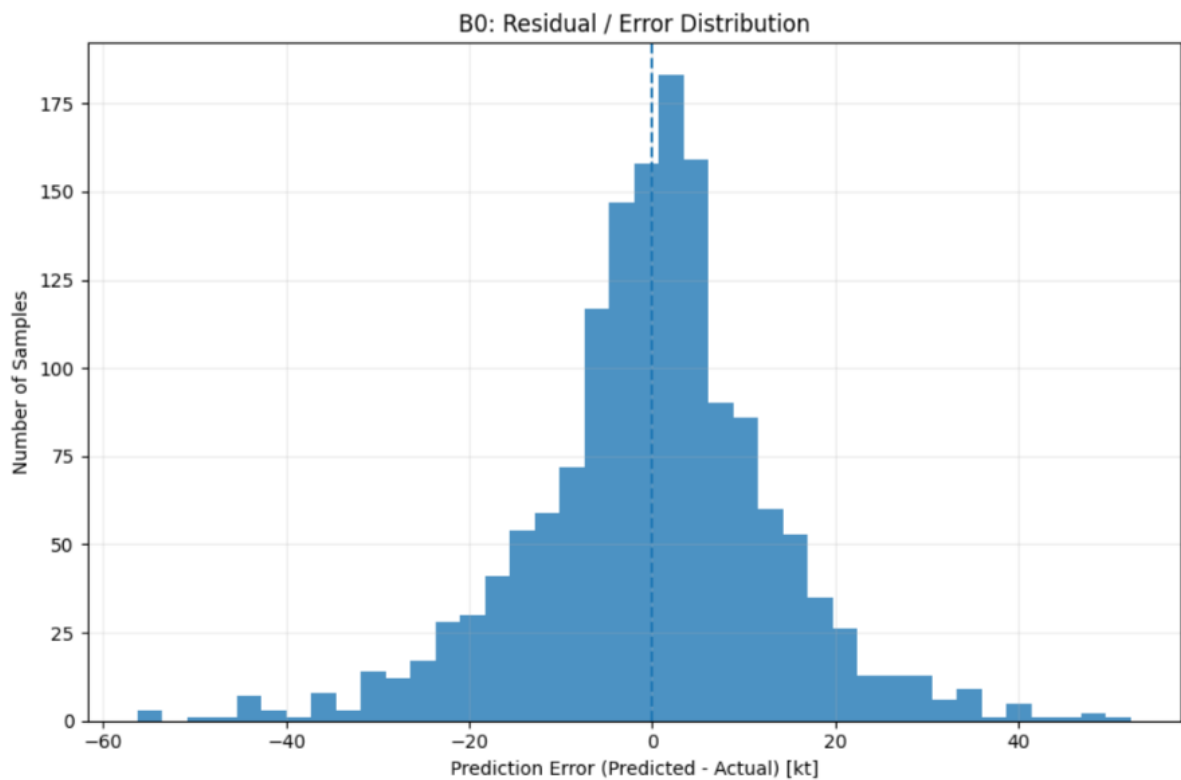


Figure 4. Residuals are predicted minus actual. The distribution is centered near zero; median absolute error is 6.8476 kt and maximum absolute error is 56.2904 kt.

10. Derived Intensity Classification

Thresholds: Weak <34 kt; Moderate 34–63 kt; Strong 64–82 kt; Very Strong ≥83 kt. Accuracy = 68.56%; Macro Precision = 0.6286; Macro Recall = 0.6403; Macro-F1 = 0.6324. A small error around a threshold can change the category even when the wind-speed estimate is numerically close.

11. Safety / Intensity-Wise Analysis

Class	Samples	MAE	Precision	Recall	F1
Weak	619	5.98	0.7545	0.8191	0.7854
Moderate	574	10.77	0.6714	0.5767	0.6204
Strong	127	14.05	0.3399	0.4094	0.3714
Very Strong	213	15.97	0.7488	0.7559	0.7523

Strong is the main weakness: recall 40.94%. Very Strong recall is 75.59%, with 161 of 213 cases correctly classified. These class-wise results are more informative than overall accuracy alone.

12. Confusion Matrix

Actual \ Predicted	Weak	Moderate	Strong	Very Strong
Weak	507	111	1	0
Moderate	163	331	65	15
Strong	2	34	52	39
Very Strong	0	17	35	161

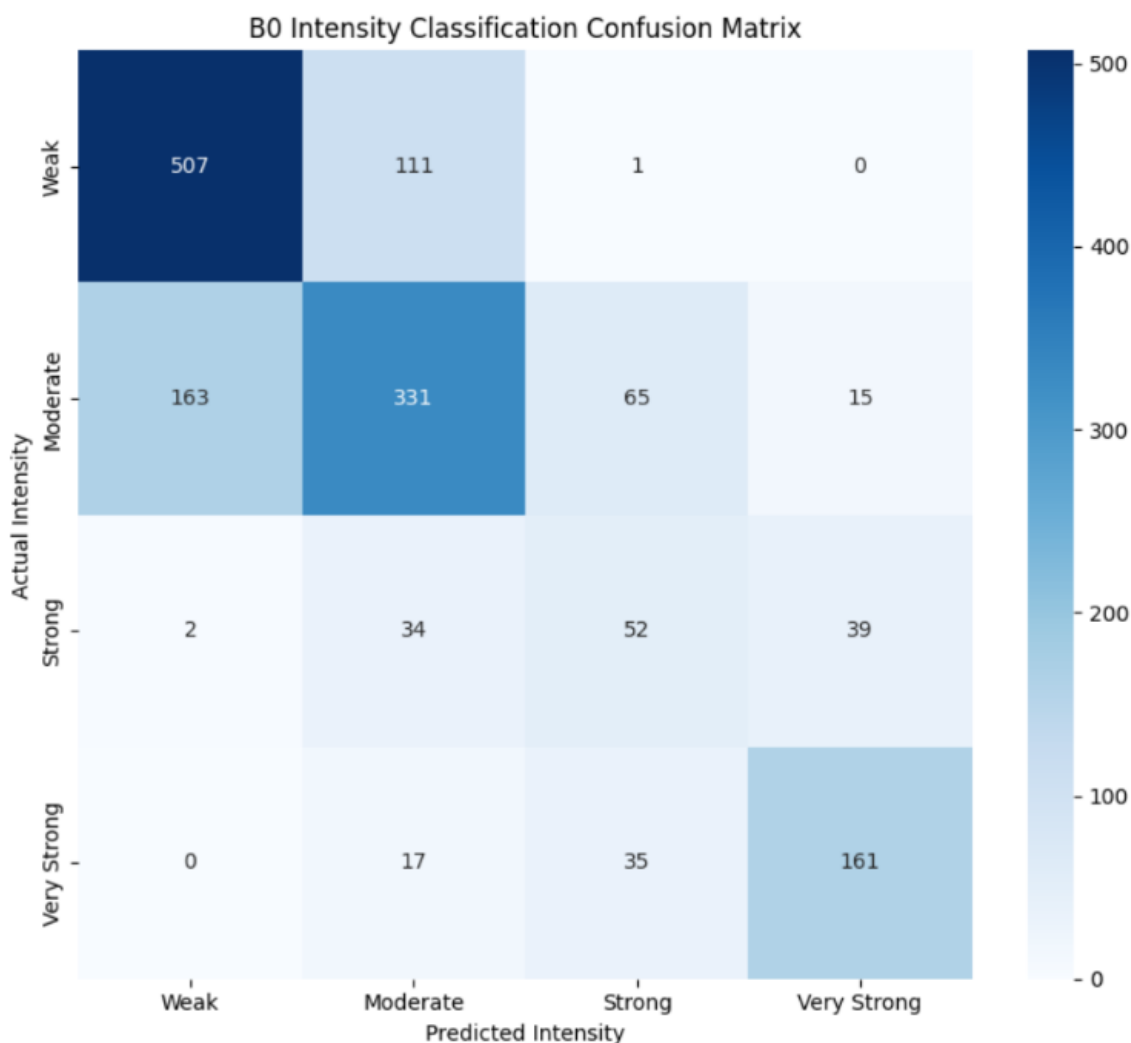


Figure 5. Classification confusion matrix. Most mistakes occur between neighboring intensity categories.

13. Why Accuracy Alone Does Not Define Model Quality

The direct task is regression, so MAE/RMSE/ R^2 are primary. Bias is almost zero. $R^2 = 0.7737$ indicates substantial explanatory power. Hard thresholds at 34, 64 and 83 kt create abrupt class changes. The test classes are imbalanced, so Macro-F1 and per-class recall are necessary. Very Strong recall of 75.59% is a useful safety signal, although it is not sufficient for autonomous safety-critical deployment.

14. Limitations and Next Development Direction

The dataset is historical and not India-specific. B0 predicts wind speed from an individual image and does not yet predict future track or future intensity evolution. Strong-category performance and the training/validation generalization gap are the main improvement targets. B0 should remain the baseline for fair comparison.

15. Conclusion

EfficientNet-B0 learned a meaningful relationship between four-channel satellite imagery and cyclone wind speed. The final held-out performance—MAE 9.83 kt, RMSE 13.45 kt, R^2 0.774 and near-zero bias—establishes a credible baseline. The 68.56% derived classification accuracy identifies a specific class-separation problem, especially around Strong, rather than proving that the regression model has learned nothing. The next stage should improve class separation while preserving continuous wind-speed accuracy.